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Toward Controllable Generative Design: A Conceptual Design Generation Approach Leveraging the Function–Behavior–Structure Ontology and Large Language Models

Recent research in the field of design engineering is primarily focusing on using AI technologies such as Large Language Models (LLMs) to assist early-stage design. The engineer or designer can use LLMs to explore, validate, and compare thousands of generated conceptual stimuli and make final choices. This was seen as a significant stride in advancing the status of the generative approach in computer-aided design. However, it is often difficult to instruct LLMs to obtain novel conceptual solutions and requirement-compliant in real design tasks, due to the lack of transparency and insufficient controllability of LLMs. This study presents an approach to leverage LLMs to infer Function–Behavior–Structure (FBS) ontology for high-quality design concepts. Prompting design based on the FBS model decomposes the design task into three sub-tasks including functional, behavioral, and structural reasoning. In each sub-task, prompting templates and specification signifiers are specified to guide the LLMs to generate concepts. User can determine the selected concepts by judging and evaluating the generated function–structure pairs. A comparative experiment has been conducted to evaluate the concept generation approach. According to the concept evaluation results, our approach achieves the highest scores in concept evaluation, and the generated concepts are more novel, useful, functional, and low cost compared to the baseline. [DOI: 10.1115/1.4065562]

Keywords: generative design, conceptual design, large language model, FBS model, creativity and concept generation, machine learning

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1 Introduction

Understanding and analyzing design requirements is a common challenge in design. Designers are often required to explore a range of potential ideas before choosing the appropriate and innovative solution to implement. In this process, early conceptualization plays an important role in determining the outcome of the design. Good concepts often result in impressive design solutions [1]. However, the quality of concepts is often influenced by the cognitive differences between individuals [2]. This has led researchers to consider how they can help designers overcome these cognitive limitations and accelerate good design.

Benefiting from the rapid development of machine learning techniques, large language models (LLMs) have gained popularity as powerful tools for a variety of natural language processing (NLP) tasks including language translation, text classification, and Q&A systems [3]. The current state-of-the-art large language models such as GPT-3 and ChatGPT even demonstrate a certain level of intelligence on some specific generative tasks [4]. LLM-based design has emerged as a novel paradigm for human–machine collaborative design [5]. LLMs can help a designer generate verbal design solutions or retrieve and interpret unfamiliar domain knowledge related to the design problem [6].

Existing LLM-based generative conceptual design methods primarily rely on knowledge-based approaches, requiring fine-tuning of the model with pre-collected design knowledge [7]. Using these methods, designers input initial fixed keywords prompt as a design requirement, and the LLM can generate potential design solutions with reasonable competence. However, it is still challenging to control LLMs to generate professional results for a specific domain [8]. The lack of transparency and insufficient controllability of LLMs can limit their effectiveness when assisting generative conceptual design [9]. The direct generation with prompts made by designers for conceptual design lacks theoretical guidance and a comprehensive understanding of design requirements. The results may lead to generating concepts that are potentially mismatched or sounding plausible but false [10,11].

To address these limitations, this study proposes a novel approach to generative conceptual design. The approach provides a task-decomposition solution from vague requirements to potential design structures. Inspired by the Function–Behavior–Structure (FBS) model, the approach contains function, behavior, and structure-related prompt templates to decompose design problems and guide LLM-aided ideation in three steps. The output of each step becomes the input for the next, aiding in aggregating gains per step and embedding the selection preferences of human designers at each stage. To evaluate the effectiveness of the proposed approach, a conceptual design experiment was conducted. The task-decomposed generative design method was compared to the knowledge-based generative method with fixed prompting, free-form prompting, and a blank setting method with a non-AI approach. The results indicate that the ideas generated by the proposed method are more novel, useful, functional, and low cost compared with other approaches.

2 Related Work

This study builds on research from the disciplines of design and computer science, including conceptual design, design methodology, machine learning models, and generative design.

2.1 Conceptual Design. Conceptual design is an essential phase of the product development process in which ideas are generated and transformed into a preliminary design [12]. It is also an integral part of creative industries such as engineering design, product development, and architecture. Conceptual design consists of an ordered, organizable, and targeted series of design activities from the analysis of user requirements to the generation of a conceptual product [13]. Pahl and Beitz proposed the concept of a design cycle consisting of four phases: task clarification, conceptual design, embodiment design, and detail design [14]. This framework has been widely adopted in engineering design and product development. While a well-structured process seems to lead to greater design success, an overly structured and rigid approach does not effectively drive creative design. To address this issue, Childs et al. [15] propose the creativity diamond framework, which can be used to prompt and assist in the selection of tools or approaches to be used for innovative tasks in a creative environment. By generating and refining ideas in a systematic and collaborative way, designers can create innovative and successful products that meet the needs of their users.

2.2 Inspirational Stimuli in Design. In the early design stages, the novelty, variety, and usability of conceptual design ideas are essential for designers to explore innovative and creative design opportunities [16]. However, stereotypical thinking and cognitive limitations in different individuals may restrict the ability to generate creative ideas. To stimulate designers' creativity and support design decisions, a number of studies have developed approaches to support conceptual design ideation, such as traditional creativity tools (e.g., brainstorming [17], TRIZ [18], and analogical reasoning [19]) and stimuli-based computer techniques (e.g., WikiLink [20] and digital sketching [21]). These tools can generate text or visual stimuli to provide design concepts to inspire designers and engineers. In addition, stimuli distance affects creative design to different degrees [22]. Far inspirational stimuli enhance the novelty of the design, while near inspirational stimuli tend to increase the usability of the design [23]. Designers can engage in the identification, evaluation, and utilization of highly favorable inspirational stimuli, instead of relying on random encounters with an unlimited number of stimuli [24].

2.3 Function–Behavior–Structure: A Task-Decomposed Model. To discover the regularities of conceptual design and represent the conceptual design process in a general way, Gero [25] presented the FBS model which decomposes the design task as a hierarchical system of functions, behaviors, and structures. Although FBS is not the only way to guide human's thinking about design, it is a classic task-decomposed model in design thinking and verified to be effective in design practice [25]. It provides a task-decomposition solution from vague requirements to potential structures. The FBS model has had a profound impact on the Design Theory and Methodology (DTM) community and has been widely implemented in various design domains, not limited to mechanical engineering, software engineering, and architecture. For example, Chakrabarti [26] used the FBS model to develop a computational approach for exploring design spaces in mechanical engineering. FBS model can help designers identify and evaluate alternative design solutions more effectively. Hamraz and Clarkson [27] explored the use of FBS to improve support for engineering

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change management and industrial evaluation. The FBS model's hierarchical structure and emphasis on task decomposition provide a flexible and adaptable framework to guide the design process and support effective decision-making [25]. Although the FBS model has proven to be a valuable tool for representing and analyzing complex design problems across multiple domains, the ability to capture inspirational concepts and explore potential structures depends on the knowledge acquisition of the individual using the model [28], which results in no guarantee of effective access to high-quality design solution.

2.4 Large Language Model–Based Generative Conceptual Design. Large language models (LLMs) have become a popular approach for natural language processing tasks [3]. These models are trained on large amounts of textual data and can then be fine-tuned on smaller task-specific datasets to achieve high levels of performance in a variety of natural language processing tasks [29]. One of the earliest LLMs was a deep-learning model called word2vec [30]. It was trained on a large corpus of text data and was able to learn word embeddings that capture the semantic and syntactic relationships between words. The success of word2vec led to the development of other pre-training approaches such as GloVe [31], ELMo [32], and BERT [33]. Benefiting from advanced Transformer architecture, open-source pre-trained LLMs, such as Llama 2 [34], Flan-t5 [35], and BLOOMZ [36], exhibit robust and powerful generative capabilities. Their conditional and generative information is more aligned, and the advantages of training efficiency and engineering realization are more pronounced than those of encoder-only and encode–decode LLMs.

In decoder-only LLMs, Generative Pre-trained Transformers (GPTs) introduced by OpenAI [37] have achieved state-of-the-art results in several natural language processing tasks including language modeling, sentiment analysis, and text completion. The GPT series includes several models such as GPT-2 [38], GPT-3 [39], and ChatGPT [40], each with an increasing number of parameters and higher performance in natural language processing tasks. GPT-3 and ChatGPT do not require fine-tuning and can achieve high-quality content generation given very few (few-shot learning) or even no examples (zero-shot learning).

Benefiting from LLMs' retrieval capacities that can infer external knowledge, LLM-based generative design is emerging as a new paradigm in innovative design. It's potential for reasoning design requirements and representing conceptual solutions has garnered significant attention from researchers. It involves defining a design problem by using descriptive prompts and allowing a machine learning algorithm to reason the solution space and come up with potential solutions. The algorithms learn from a dataset of existing designs or prototypes and generate new designs that meet the given specifications. Zhu and Luo [41] propose an approach for generating design concepts by using GPT2. It requires fine-tuning the model using a pre-collected design knowledge dataset or providing the model with some concrete examples to implement small sample learning before using it to generate design concepts. Ma et al. extended previous work against the human crowdsourcing baseline by evaluating LLMs-generated design solutions using computational and expert metrics [42]. Makatura et al. explored the workflow of LLMs applied in design, design tools, manufacturing, and the performance of computational design, and pointed out the utility of LLMs under design and manufacturing contexts [43].

However, these works still lack clarity on how to guide LLMs to efficiently generate ideas that support the early stages of design. Current general decoder-only LLMs are primarily based on prompt engineering. Without limiting the prompt of reasoning content, many of the generated results will be ambiguous and difficult to use. The excessive number of invalid answers generated by LLMs may even lead to an additional cognitive burden for designers. Despite more powerful models such as GPT-3 and ChatGPT being developed, which can perform zero-shot inference tasks through prompting, it remains challenging to obtain convincing results in the specific domain due to the model's lack of transparency and insufficient controllability.

Our literature review suggests that previous research work has aimed to develop a large language model-based approach to generate conceptual solutions to support designer concept design ideation. However, prompting without guidance is not sufficient for controllable content generation. It is common that AI models deviate from the intent of original generative tasks and infer text on irrelevant topics. Moreover, preparing domain-specific knowledge as a fine-tuning dataset is time-consuming, and human-in-the-loop interactions during the process of the generation are not considered. It seems promising to propose an LLM-based generative design approach that is controllable and transparent to generate design-related concepts.

3 A Function–Behavior–Structure–Based Task-Decomposed Framework for Generative Design

3.1 Methodology Overview. We proposed a framework for explicitly prompting the LLM to decompose responses into the FBS ontological framework, which has been illustrated in Fig. 1. The implementation of the framework has been made open-source.[2] Prompt templates and specification signifiers are used to guide the model in reasoning about the design problem correctly and controllably at the different generative stages. The sub-prompt templates correspond to the three mapping stages of FBS, including requirement to function ($R \rightarrow F$), function to behavior ($F \rightarrow Be$), and behavior to structure ($Be \rightarrow S$). In contrast to the previous LLM-based generative design, a designer can filter the generated content at each stage of the generation process, thus incorporating human interpretation of the design representation. To help designers better leverage the generated descriptions, understanding the relationship between functions and structures in the FBS model is necessary. Therefore, the functional contents generated in $R \rightarrow F$ Mapping and the corresponding structural contents generated in $Be \rightarrow S$ Mapping are organized into paired "function–structure" stimuli. The designer can evaluate novel and reasonable design structure suggestions based on the potential functions needed for a design requirement. The generated stimuli of concepts are the result of human-computer collaboration, which enables a designer to use them for creative and reasonable conceptual design in the phase of ideation.

3.2 Function–Behavior–Structure Transformation Process and Ontological Components. In the FBS transformation process, A design problem***PT*** is usually decomposed step-by-step into three ontological components: function, behavior, and structure [44]. The FBS framework and the process are shown in Fig. 2. Function refers to the tasks or goals that the system is supposed to achieve. These are the ultimate purposes of the system design, describing what it is supposed to do. Structure refers to the physical composition or configuration of the system, including its components and the relationships between them. The structure defines the physical attributes of the system, such as shape, size, materials, and the layout of its components, essentially representing the tangible part of the system. Behavior describes how the system accomplishes its functions. It covers the dynamic characteristics of the system, including how it responds to different inputs and environmental conditions. Behavior can be further divided into expected behavior (Be), which is anticipated during the design phase, and actual behavior (Bs), which is exhibited during structure operation. The comparison between Be and Bs helps identifying discrepancies between the designed and actual performances of the system. This evaluation is crucial for iterative design processes, where such discrepancies can inform adjustments and improvements in the design

[2] https://github.com/zju-d3/A-Conceptual-Design-Generation-Approach-Leveraging-the-FBS-Ontology-and-Large-Language-Models.git

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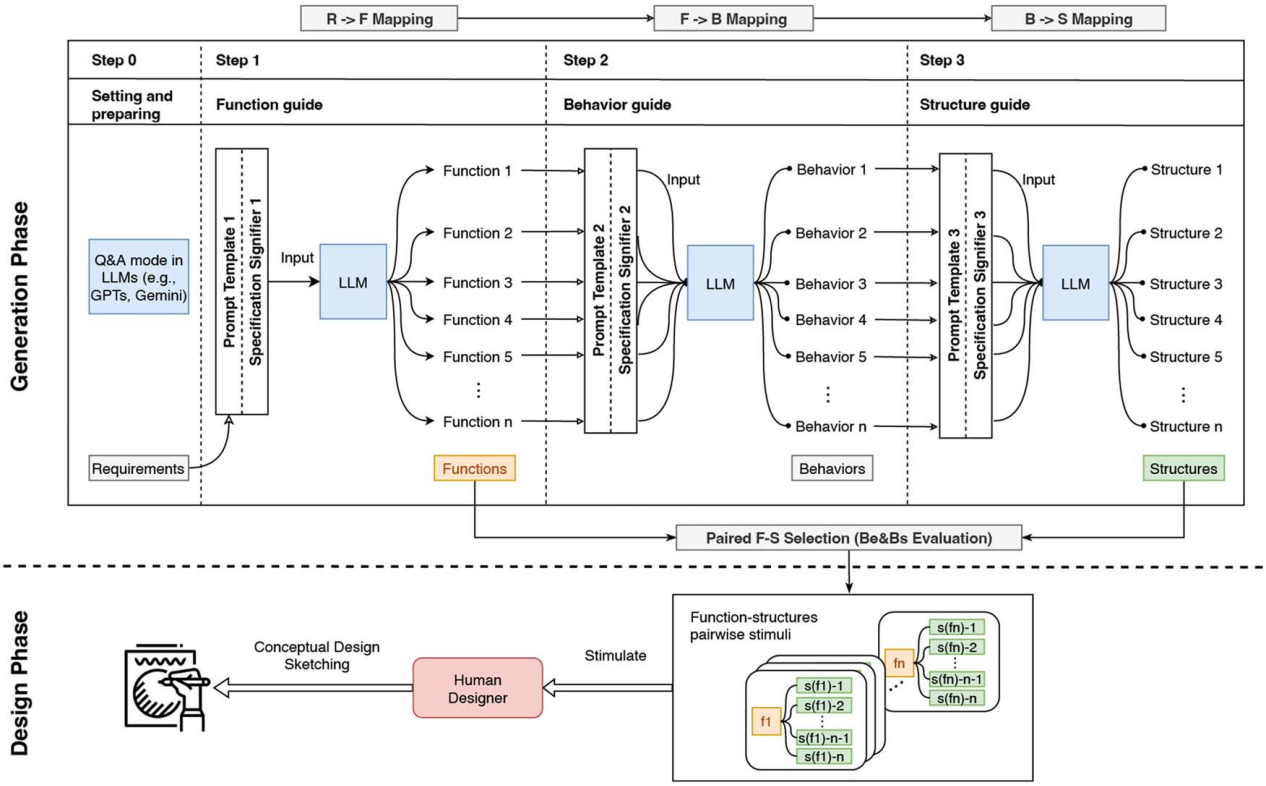


Fig. 1 The FBS-based task-decomposed approach for generative conceptual design. In generation phase, designer initially inputs requirements to LLM. The LLM will inference design concepts from Function, Behavior, to Structure through FBS process, and designer can filter the generated content at each stage of the generation process. Once LLM finishes the reasoning, designer will choose the functions that they think to be good, and the structures that can meet the corresponding function. Then, the paired Function–Structure will be used as design stimuli to inspire designer to create conceptual design sketch in design phase.

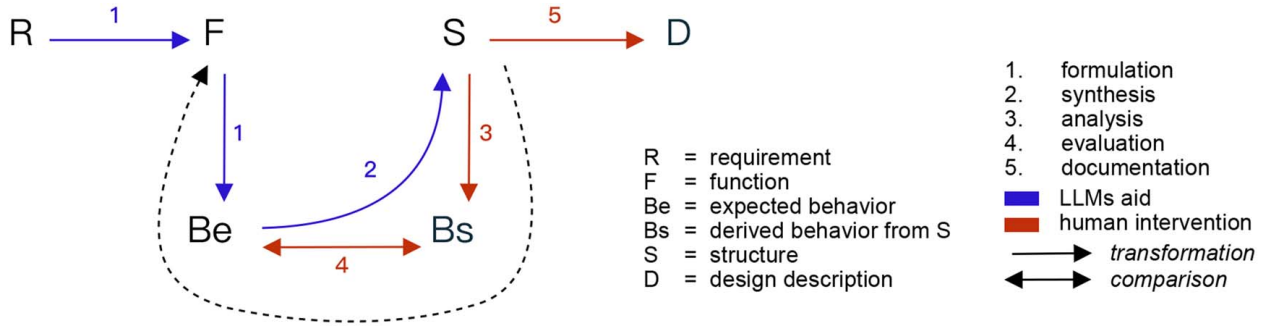


Fig. 2 The scope of intervention of LLM and human in the FBS transformation process

to better align the actual behavior with the intended functions and expected behavior.

Inspired by the FBS model, the adopted approach decomposes an abstract and fuzzy design requirement into three representable parts, including formulating a function space by transforming design requirements ($R \rightarrow F$), transforming functions into an expected behavior space ($F \rightarrow Be$), and generating structures based on the expected behaviors ($Be \rightarrow S$). Similar to other design support models, the output of FBS model relies on human preferences, selection, and evaluation. Therefore, in this study, we only integrate the formulation and synthesis stages of the FBS ontology into the LLM reasoning. The analysis stage from the structure to the behavior derived from the structure ($S \rightarrow Bs$), the evaluation stage between the expected behavior and the derived behavior ($Be \leftrightarrow Bs$), and the documentation stage to produce a design description based on the structures are performed by the human designer. Figure 2 illustrates the scope of LLM and human intervention in the FBS transformation process.

It is worth noting that the FBS evaluation describes the consistency measurement between Be and Bs, which reflects the rationality between structure and function. Clearly, it is not necessary for the behavior component to be represented as an intermediate mediator in the design description. In contrast, the function and the corresponding structure reflect possible design paths for the requirement. Therefore, the potential structures and their corresponding function were organized into "function–structure" pairs as the inspirational stimuli.

3.3 Large Language Model Configuration. Today's state-of-the-art LLMs differ in terms of their training strategies, model architectures, and use cases. With the advent of GPT-3,

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more and more companies are opting for closed-source, decoder-only LLMs. Although decoder-only LLMs lack sufficient controllability and transparency, their impressive reasoning capabilities and applicability challenge the dominance of custom-trained encoder-decoder LLMs such as BERT. In addition, the generative performance of LLMs is highly dependent on the components of pre-trained data. As the foundation of the amazing capabilities of LLMs, the quality, quantity, and diversity of pre-training data significantly affect the generated contents of LLMs. Pre-trained data provide language models with rich lexical knowledge, syntactic, and semantic understanding, and the ability to recognize context and generate coherent responses. Therefore, we recommend selecting decode-only LLM with carefully curated pre-training data, such as GPT-3 [39], ChatGPT [40], LLaMA [45], and Bard [46], to ensure a comprehensive reflection of human knowledge.

An LLM features various application modes such as sentence completion, summaries and keywords, Q&A, and code explanation. Given the need for appropriate reasoning within the context of design task decomposition, we specified Q&A as the reasoning mode for this study. We input a design question or sub-question and expect the language model to provide novel, diverse, and reasonable answers. In addition, existing LLMs have demonstrated robust performance in zero-shot learning [47]. We do not implement any further fine-tuning or few-shot learning for LLMs, which can reduce additional time and labor consumption associated with collecting and constructin***PT***g design datasets.

3.4 Prompt Template. The generation of LLMs is usually performed with a text prompt. By creating suitable prompts, we can manipulate the behavior of the model to enable it to predict the desired output. Prompts are usually created manually by users (see Supplemental Material available in the Supplemental Materials on the ASME Digital Collection).

Given an inappropriate prompt, a user may not be able to retrieve the facts that an LLM does know, and models only provide a lower-bound estimate of the knowledge contained. In order to allow LLMs to understand the input and to make reasonable inferences easily and correctly, designing a prompt template is worth considering [48,49]. A prompt template is a direct***GC***ive text framework designed to facilitate the generation or elicitation of sp$\boldsymbol{M}_{ptl}$ecific types of responses, co*x*ntent, or data. Its applications and formats vary across different contexts [50]. For instance, in the field of NLP, prompt templates are employed to guide AI models ix*n* interpreting and responding to input queries or tasks. These templates typically comprise predefined instructions, questions, or statements, aimed at steering the model toward producing a targeted output. A prompt template can be represented as a sentence wrapper for organizing keywords or phrases. The instance-wrapped prompt will be fed into an LLM to generate the corresponding content, as shown by Eq. (1):

$$\boldsymbol{GC} = \boldsymbol{M}_{ptl}(\boldsymbol{PT}(x)) \tag{1}$$

where x refers to the keywords or phrases to be input, $\boldsymbol{PT}$ represents the prompt template used to wrap the input content, $\boldsymbol{M}_{ptl}$ represents the LLM used to reason the instance prompt, and $\boldsymbol{GC}$ represents the generated content. For example, ChatGPT can be regarded as an instance of $\boldsymbol{M}_{ptl}$, and a simple $\boldsymbol{PT}$ can be represented as "What is ($\underline{x}$)," where [item_name] ∈ {"gear," "agitator," and"torque"}. The designer can choose an item from [item_name], such as "gear," and fill it into the placeholder ($\underline{x}$). For example, a completed and organized prompt could be "What is gear," to which ChatGPT might generate the response, "a gear refers to a toothed mechanical component that meshes with other gears to transmit motion and power." To obtain answers for different objects, the user only needs to modify the keywords without altering the overall linguistic description to get the results they desired.

3.5 Specification Signifier. FBS-based design analysis is a complex quest that contains several sub-tasks. Directly defining a uniform prompt template for FBS-based AI generative design is challenging. An intuitive solution to this problem is to decompose a quest prompt into different sub-prompts [51], an approach that has shown LLM-based decomposed prompting approach can better address large tasks requiring multiple predictions for a single sample. However, unlike prompt design based on fine-tuning or few-shot learning, zero-shot learning-based LLMs have a limited ability to understand the abstract description of design tasks. Analyzing possible forms of task specification from vague questions is difficult. To address this issue, specification signifiers [52] have been created for each sub-prompt template. A specification signifier can be a keyword or phrase that specifies the information LLMs generate related to the request target. It can be a noun, phrase, or verb that describes the prompt, considering the contextual voices. The specification signifier regulates reasonable model generation for each sub-task. It can effectively reduce the difficulty of a zero-shot learning-based LLM in understanding prompts and make the generated content more reliable.

4 Experiment and Evaluation

To evaluate our method's performance against other generative design methods in supporting conceptual design ideation, we organized a comparative experiment and chose ChatGPT (model: gpt-3.5-turbo-0125; knowledge cutoff: Up to September 2021), the most representative LLM, for this experiment. The parameters of the GPT in this experiment, such as seed and temperature, were set to default. The overview of the experiment is shown in Fig. 3.

4.1 Prompting Design. Since LLMs are insensitive to different prompts that preserve the semantic essence of the original text, it is not strictly necessary to recover every single detail perfectly from the original intention [53]. The generated result can satisfy the user's original requirements if given a prompt that conveys the necessary level of semantic precision or intent [53,54]. To reduce the bias introduced by long tokens and word semantics, in the experiment, we specified three simple interrogatives as well as three keywords of FBS ontologies as the prompt templates and specification signifiers for each sub-task, as shown in Table 1. The specification signifiers for each template are denoted as "function," "afford," and "structure," allowing the model to efficiently capture FBS-related knowledge within its potential space in the zero-shot prompt setting. The placeholders in the templates, such as "[RQM]," "[FCN]," and "[BEH]," indicate the information that needs to be filled in by the user. It should be noted that we do not use the word "behavior" as the specification signifier in the prompt template transitioning from function to behavior. In Gero's model, behavior represents the behavioral logic of interaction that the product structure may afford. For example, the four-way scroll key on a cellphone can afford behaviors such as swiping through the information on the screen. Behavior is not the carrier from which the function emanates, but the logical trigger that causes the function to occur. At the level of natural language understanding, if "behavior" is used to prompt the model to reason about behavior, the model tends to interpret "behavior" as an action that might occur in a generic context. Considering the model's limited reasoning capacity, we believe that using the word "afford" makes it easier for the model to understand this ontology as an interaction logic that may dictate a specific function.

4.2 Constructing the Comparative Experiment. We defined a conceptual design challenge where the participants needed to sketch a leg sports rehabilitation device according to the generated contexts by ChatGPT. The leg sports rehabilitation device, a typical and important medical equipment design in mechanical

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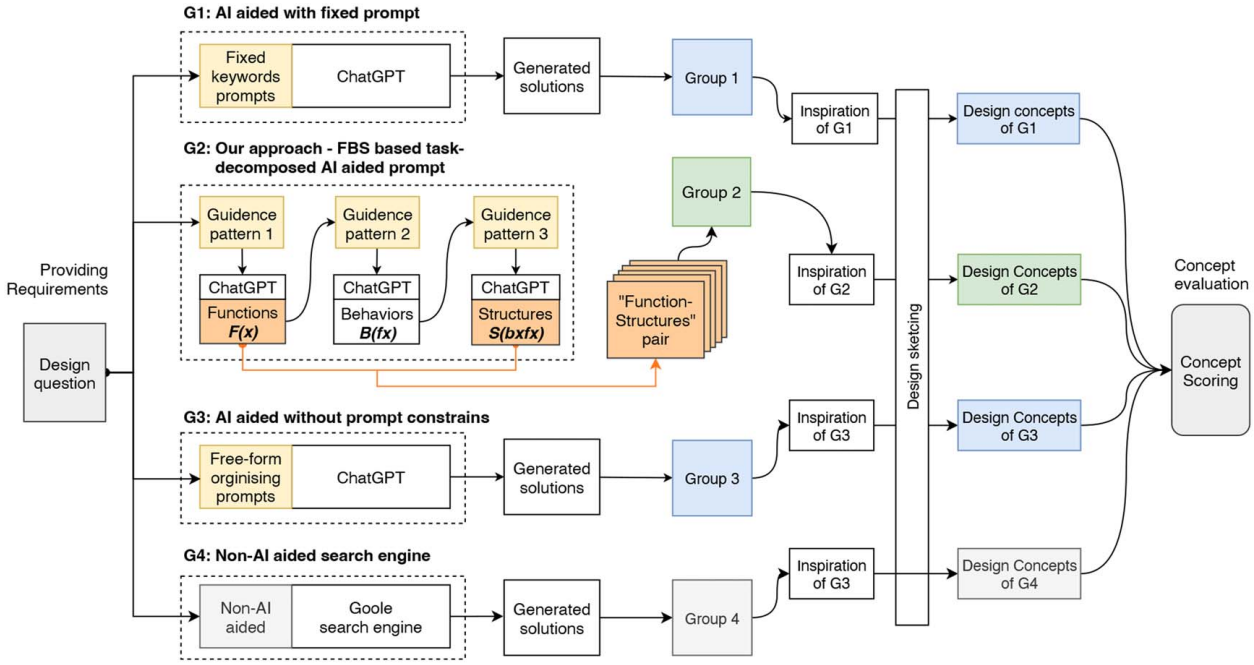


Fig. 3 The comparative experiment of FBS-based concept generation

Table 1 The prompt templates and specification signifiers of FBS-based ChatGPT for generative conceptual design

Stages of sub-task	Specification signifiers	Placeholder	Prompt templates
Requirement → Function	'function'	[RQM]	"What functions should a (an) [RQM] have?"
Function → Behaviour	'afford'	[FCN]	"What kind of design can afford [FCN]?"
Behaviour → Structure	'structure'	[BEH]	"What design structures can achieve [BEH]?"

engineering, has significant implications for patient recovery. The quality of sketching can be related to the performance of the individual designer's hand-drawing skills [55]. Professionally trained designers and those from interdisciplinary backgrounds demonstrate varying levels of hand-drawing ability. However, this does not mean that people with poor hand-drawing skills have equally poor design thinking. To reduce bias in the process of generating text stimuli for conceptual design sketches, 80 designers with similar hand-drawing skills were recruited, including 60 novice designers with 3–4 years of undergraduate design education and 20 professional designers with over 8 years of design experience.

In this study, 80 participants were randomly and evenly assigned to four groups to investigate the impact of different methods of design proposal generation, where each group had the same number of novices (15 participants) and professional designers (five participants). Group 1(G1), group 2(G2), and group 3 (G3) are based on different AI-aided approaches, and group 4 (G4) is a baseline group without AI-aided. G2 is the experimental group which utilizes the FBS templates to guide the design requirement input. G1 and G3 are two control groups. In G1, participants generate design proposals based on fixed prompts proposed by Zhu and Luo [41] which was fine-tuned by existing design with "keyword-description" pairs from the design awards database. In G3, participants freely prompted GPT questions based on their understanding of the design requirements and then generated design proposals. In addition, G4 that participants access a search engine with non-AI driven force was introduced to understand whether using an LLM during the concept generation phase is even useful. By comparing the performance of these four groups, we aim to provide insights into the effectiveness of different design proposal generation methods. The experiment was conducted in the same room, at the same time, with all guiding banners, posters, and decorations masked to ensure reliability, and the whole process was limited to 30 min. For G4, Participants were instructed to use the Google search engine only and were restricted to use any AI tools. For G1–G3, the participants were provided with a computer setup with ChatGPT and were familiarized with how to complete the design generation using their group's method before the experiment. All participants were asked to first generate design solutions using ChatGPT, then complete the design idea and sketch solutions. In addition, they were forbidden to use any other software or search engine or to input anything into ChatGPT beyond what was requested. An example of the concept generation from each group is shown in Fig. 4.

Traditionally, the method used by the G1 is known as knowledge-based generative design with fixed prompt based on few-shot prompting, requiring several inputs of keyword-description pairs from existing design databases. In this case, we collected the keyword-description pairs from the industrial design products in Red-dot Awards. The participants were asked to enter "Leg, sports rehabilitation equipment" as the fixed input keywords into ChatGPT. With the keyword pair, the ChatGPT would generate a design description as instructed. If participants were not satisfied with the response, they could regenerate multiple times to obtain different results. However, participants were asked not to modify, add to, or delete the original input. These suggestions were then used as the stimulations to complete design ideas and sketches.

The G2 serves as the experimental group in this study, where participants follow the FBS prompt templates to generate design proposals. The FBS prompt templates provide a structured approach to generating design proposals that include three stages: requirements-to-function, function-to-behavior, and behavior-to-

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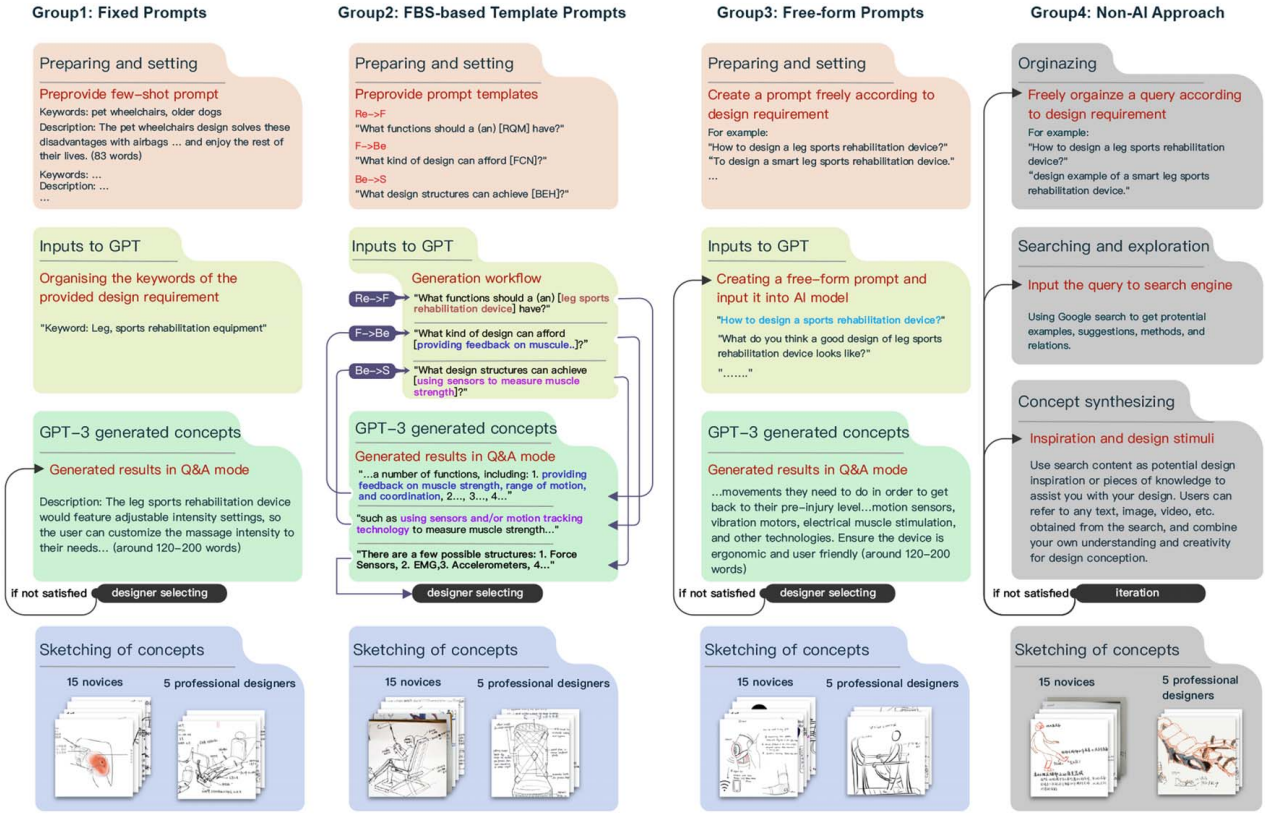


Fig. 4 An instance of the generation process from each group. Group1 uses fixed prompts providing keywords-description to generate concepts; Group2 is our approach by integrating FBS and ChatGPT; Group3 is the free-form based LLM group that humans can input any prompts to generate concepts according to their understanding of requirements; Group4 is a non-AI-aided group that user use traditionally search engine to get ideas and the generation process doesn't involve LLM.

structure. In the requirements-to-function stage, we embed the requirements text into the first prompt template, which is expressed as "What functions should a (an) leg sports rehabilitation device have?". The participants were required to select one or more functional phrases or words from the output which are considered to be reasonable. These words would be recorded as input keywords for the second prompt template. In the function-to-behavior stage, participants needed to select a recorded functional word to embed into the second prompt template. Same as in the first stage, participants were allowed to select and record one or more words from the output that they considered reasonable behavior. In the behavior-to-structure stage, the recorded behavioral words were used directly in the third prompt template to generate a structural solution. These solutions, combined with the functional word entered in the second stage, serve as the "function–structure" stimuli. In Fig. 4, for example, structures such as "force sensors," "EMG," and "accelerometer" can be paired with a function "providing feedback on muscle strength". It is easy to associate the use of EMG with the ability to calculate muscle strength feedback. Participants can repeat the process from the second to the third stage to obtain more structured suggestions for different functions, using these pairwise sources as stimuli to complete design concepts and sketches.

The G3 serves as the control group, where participants were asked to input a question or task description related to designing a leg sports rehabilitation device into ChatGPT to receive generated suggestions. Participants were free to organize the question or task description based on their understanding and design concepts, and input it into the ChatGPT dialogue box to obtain AI-generated design proposals or potential design suggestions. If participants were not satisfied with the generated results, they could regenerate multiple times to obtain different results. These suggestions would be used as stimuli to complete design ideas and sketches. As a result, G3 has no guidance and structure supporting participants in design concepts in the generation process. It provides a baseline outcome without AI assistance for the G1 and G2 groups.

In the G4 group, participants were allowed to use the Google search engine to explore potential solutions. They were allowed to enter whatever they wanted to search for as many times as they wanted based on their understanding of the design task. The input can be a complete sentence or a textual concept fragment. The search results can be any text, images, audio, and video. They were allowed to record and save any search results and to complete the sketch based on these results. If they were not satisfied with the search feedback, they could adjust the inputs at will to get new results. Participants were banned from using any AI-related tools, such as Q&A systems with LLM integration and AI-embedded search engines, in the whole process.

After creating the design sketch, we conducted individual interviews with a minimum of two novice and two professional designers from each group. The objective of these interviews was to gather valuable insights and experiences from the designers regarding the design process and the generated results. The information obtained from these interviews allowed us to gain a comprehensive understanding of the advantages and disadvantages of the three methods. The results are reported in Sec. 4.2.

5 Evaluation

5.1 Concept Scoring. Conceptual design evaluation plays a crucial part in the early stages of product design as it has a major impact on the downstream production process and the success of the product being developed. Pugh's decision metrics is a widely accepted method for conceptual design evaluation [56]. It helps designers make the right decisions during the concept selection phase. This method is roughly divided into

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Table 2 Rating generated concepts with reference concept

Relative Performance	Rating
Much worse than reference concept	1
Worse than reference concept	2
Same as reference concept	3
Better than reference concept	4
Much Better than reference concept	5

three steps: selecting an evaluation reference, specifying criteria, and rating.

An effective and common way to validate the generated output is by using existing design objects from design awards as the references for comparison [57,58]. Since a project that wins an international design award represents a high level of quality and novelty in its specific division [59], we selected a limb rehabilitation device called "DANU Sports" from the Red-dot awards (red-dot.org) as the reference concept. The reference was awarded a gold medal in conceptual design and is also the only one awarded in that year in the medical equipment division. It integrates a wearable leg rehabilitation system with a sock, using breathable and antibacterial fabric to reduce the chance of infection after injury. Various sensors collect athletes' movement data to monitor lower limb loads, analyze gait, quantify fatigue, and track rehabilitation progress.

For the choice of criteria, we referred to the criteria setting of creative design in Ulrich and Eppinger's concept scoring [60], which is an improvement of Pugh's method. In this project, novelty, feasibility, usability, functionality, and cost were considered. All five criteria were assigned the same weight of 0.2. We did not set or specify additional weights for these criteria to reduce the impact of multiple variables on the method.

80 conceptual sketches drawn by participants were randomly mixed for comparative evaluation with the reference concepts. A rating based on the 5-point Likert scale was used to examine the performance of each concept relative to the benchmark concept. Three experts, each with 10 years of design experience, were employed as the assessors. If a sketch scored higher than the reference on any criterion, it was given a higher score for that criterion and vice versa, as shown in Table 2.

5.2 Evaluation Results. Given that the aim of this study is to assess the generative performance of the method for supporting conceptual design ideation, rather than the scores of individual concepts, we calculated the ratings of novices and professional designers in each group separately, as well as the average score of each group, as shown in Table 3. C1 to C5 represent each of the five different criteria respectively. The scores of novices and professionals under each criterion are the weighted averages of their five and are used to compare with the base score. The final average scores of the groups are used for comparison with each other. Additionally, we calculated the inter-rater reliability using Fleiss' Kappa [61]. Fleiss' Kappa is a statistical measure used to assess the consistency among multiple raters when evaluating items. The value of Fleiss' Kappa ranges from −1 to +1, where −1 indicates total disagreement, 0 signifies a low level of agreement at the level of chance, and +1 denotes perfect agreement. Generally, a Fleiss' Kappa value exceeding 0.61 is considered to reflect a high degree of consistency. The alpha values of Fleiss' Kappa for each criterion are shown in Table 3. The resulting Fleiss' Kappa for the average score is 0.66, indicating substantial agreement among the raters in the scoring process.

To facilitate concept scoring, the mean and standard deviation (SD) for each group were calculated. The results, reported in Table 3, show that the average concept score of G2 is 3.31 (SD = 0.58), the highest among the four groups. The average concept score of G1 is 2.94 (SD = 0.55), which is higher than G3 (mean = 2.90, SD = 0.47) and G4 (mean = 2.63, SD = 0.53). To verify the statistical significance of the results, a One-Way ANOVA was conducted. The results revealed statistically significant differences among the groups. The analysis yielded an F-statistic of approximately 15.99, indicating a significant variance among the groups. The associated p-value was approximately 1.69×10^{-9}, strongly statistically significant differences across the groups. This low p-value ($p < 0.05$) indicates that the observed variance is unlikely to have occurred by chance, leading to the rejection of the null hypothesis of equal means across the groups.

According to the Shapiro-Wilk normality test results, the data for groups G1 to G4 are relatively consistent with a normal distribution, indicated by p-values of 0.082, 0.062, 0.072, and 0.059, respectively, all greater than the commonly used significance level of 0.05. Therefore, to determine the effectiveness of introducing LLM and FBS in improving concept generation quality, we calculated the independent t-test between G4 and G1/G3, and between G2 and G1/G3, respectively. The statistical analysis indicates statistically significant differences in G4 versus G1 (t statistic = −3.24, $p = 0.0016 < 0.05$), G4 versus G3 (t statistic = −2.90, $p = 0.0044 < 0.05$), G2 versus G1 (t statistic = 3.18, $p = 0.0019 < 0.05$), and G2 versus G3 (t statistic = 4.22, $p = 4.84 \times 10^{-5} < 0.05$). In addition, there is also significance between our method (G2) and the G4 (t statistic = 6.641, $p = 1.007 \times 10^{-9} < 0.05$). The results are shown in Fig. 5.

In addition to the results presented in Table 3, we conducted semi-structured interviews after sketching. The main questions of the interview focused on the participants' experiences using the design tool, specifically regarding usability, efficiency, and pros and cons. Each interview lasted approximately 45 min was audio-recorded, transcribed verbatim, and anonymized for analysis. Thematic analysis was used to identify common themes and variations in participant responses. Part of quotes were categorized under each

Table 3 Evaluation results of conceptual design in each group

Evaluation criteria			Design concepts									
			Group 1 (Fixed form)		Group 2 (Our approach)		Group 3 (Free form)		Group 4 (Non-AI)			
Criterion	Representation	Weight	Base	N	P	N	P	N	P	N	P	Inter-rater Fleiss' Kappa
C1	novelty	0.2	3.0	3.49	3.20	3.53	3.60	3.25	**3.73**	2.71	3.00	0.63
C2	feasibility	0.2	3.0	2.87	3.20	**3.64**	3.53	2.85	3.46	2.60	3.30	0.77
C3	usability	0.2	3.0	2.80	2.80	**3.58**	3.00	2.49	3.23	2.42	3.27	0.76
C4	functionality	0.2	3.0	2.34	2.73	**3.26**	3.20	2.40	3.07	2.42	2.73	0.71
C5	cost	0.2	3.0	3.09	2.93	3.15	**3.40**	2.73	3.18	2.44	2.60	0.79
Average Score			3.0	2.92	2.98	**3.43**	**3.35**	2.75	3.33	2.52	2.98	0.66
Weighted Average Score			–	2.94 (SD = 0.55)		**3.31** (SD = 0.58)	2.90 (SD = 0.47)		2.63 (SD = 0.53)			–

Note: Bolded scores represent the best scores on each criterion, and the average score of the group based on our approach. N = Novice designers, P = Professional designers, Weighting assigned: $N = 0.75$ (15/20), $P = 0.25$ (5/20).

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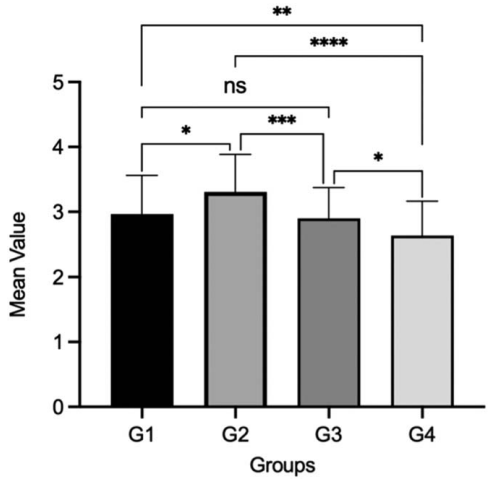


Fig. 5 Comparative analysis in t-test across four groups with indicators of statistical significance

theme to support the qualitative findings and exemplify key points, as shown in Table 4.

For the G1 group, the advantages primarily center around the tool's capacity to offer fundamental guidance with fast operation, aiding users in clarifying product requirements and collecting information at a relatively low cost. However, it is noted that the tool is confined to existing product queries, limiting user's inspiration and innovation, and may pose certain difficulties in sketching and lack guidance on appearance. In the case of the G2 group, the feedback on user experience mainly focuses on the tool's ability to provide various design inspirations, generate results that are close to feasible for specific functions and structures, and offer new ideas and solutions. This approach is controllable and practical, helping users to ideate and determine design solutions. For the G3 group, the advantages and disadvantages primarily revolve around the tool's ability to provide basic and clear answers, assisting users in clarifying problems and obtaining correct answers. However, constructing prompts challenges designer's knowledge base and conceptualization abilities. Freely inputting and accessing information allows participants to explore LLM's knowledge space in ways that they excel at. Organizing inputs, screening outputs, and judging and refining information based on the context are challenging.

6 Discussion

6.1 Discussion of Evaluation Results. By calculating the weighted overall average score for the 20 concepts in each group, G2 outperformed the other groups with the highest concept score of 3.31, exceeding G1 by 12.59%, G3 by 14.14%, and G4 by 25.86%. This finding suggests that our task-decomposed approach based on the FBS model can be considered an effective approach in the conceptual design ideation. The results of this study show that G2 achieved the highest values in each criterion. A possible explanation for the highest values in novelty may be that the paradigm based on the FBS structure splits a complex design problem into three sub-tasks which are easy for machines to infer the concepts that are unfamiliar to human designers. In addition, the observed highest values in feasibility, functionality, and usability in G2 could be attributed to the specification signifier and the prompt template. The former provides a domain-specific representation of the prompt for each sub-task and may have activated some design-specific neurons in the LLM, making it more capable of capturing information that matches the design requirements during the model reasoning [52]. The latter constrains the semantic intent of the prompt, making the model more controllable [49].

G4 was a traditional baseline group without LLM involvement. It received the lowest score we expected. This suggests that LLM assistance is indeed more effective for conceptual design compared to traditional search engines. One possible explanation is that LLMs are built-in autoregressive search engines [62]. By allowing LLMs to use cross-attention to interpret the question and each token in the document, they achieve better information search and generalization abilities than traditional dense retrieval search engines. In contrast, LLM is more like a systematic review of learned information, recommending and organizing the most relevant content to the user [63]. The results further argue that the reasoning ability of LLM can be applied to aid conceptual design. However, this kind of assistance is not just about obtaining a result through AI agents and that's it. This suggests a new perspective of LLM-supporting design that LLM-aided conceptual design moves toward guiding LLM-generated conceptual designs through design science strategies.

The interview result indicates that G1, G2, and G3 group tools have their own advantages and disadvantages. The G1 and G3 group tool provides basic guidance and helps users clarify product requirements and design direction at a low cost, but is limited to existing product ranges and may be lacking in inspiration and innovation. In contrast, the G2 group tool provides diverse design inspiration, generates practical solutions for specific functions and structures, and helps users obtain diverse design directions and ideas. Although some answers may be too brief, the generated results may have errors and deficiencies that require users to filter and optimize. Even with no restrictions on input in the G3, many of participants reported that when they input questions such as "What design plans can achieve 'Leg, sports rehabilitation equipment'?" the AI output was often too general, which made it difficult to focus on design domain-specific problems. For example, one designer wanted to learn how to design a rolling system that would allow users to adjust the tension level quickly and easily,

Table 4 Example of the interview quotes in each group

	Example of interview quotes
G1	A participant with professional experience stated that "the method can give a general design direction in a very short time. This is because I don't need to spend time to think while inputting. But the simple and rough input regulation also limits me to input some different ideas that come from my mind". Another one also mentioned that "once the generated descriptions get too specific, the conceptual design is essentially a visualization of the textual descriptions, rather than laying in the designers themselves"
G2	An expert from G2 said that "during the questioning process, I could get progressively deeper and more detailed answers from LLM. This way is more in line with my usual workflow and intuition of thinking in terms of conceptual design". In addition, this kind of approach expands designers' knowledge space and helps them quickly target potentially design structures that have functional attributes to address the needs. As an expert stated, "when given a requirement but I am unsure of how to proceed, this method can help me to quickly explore and clarify possible ideas while reducing the cost of information gathering." A novice also said that "I feel like that without this tool, I would have to go through the information on that design one by one to see what structures are available to fulfil the needs." Nonetheless, pairwise function–structure responses are brief, necessitating users to further understand and explain, and the generated results may contain certain errors and deficiencies, prompting users to screen and optimize
G3	As two participants mentioned, "I didn't know how to start organizing inputs at first, and spent a lot of time trying to figure out how to prompt AI, but ended up asked LLM in the simplest way possible—how to design a sports rehabilitation device?"

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but the AI output only introduced the definition of a conveyor belt in the industry, which was not related to adjusting the tension level. Overall, users can choose to use these tools according to their preferences and needs. The results of our experiment and individual interviews suggest the G2 group tool stands out in its ability to generate practical solutions and expand the design process's possibilities.

6.2 Discussion of Novices and Professional Designers. The aim of this study is to investigate the difference between AI and non-AI approaches, and between FBS and non-FBS approaches. Introducing both experts and novices as participants in the experiment aims to increase the reliability of simulating real design tasks (where both novices and experts can be the designers). However, comparing participants with the same background in different groups is still worth exploring.

When comparing novice or professional designers in different groups separately, the novices and professional designers of the G2 still achieved the best performance. This indicates that the proposed approach is effective across different levels of design expertise. Compared to previous methods, the FBS process decomposition model provides improvements in the conceptual output for both novice and professional designers. Specifically, compared to G1 and G3, the average score for novice designers in G2 improved by 0.51 and 0.68, while the score for professional designers improved by 0.37 and 0.02, respectively. This suggests that novice designers in our approach achieved greater improvements in design average scores than the LLM-based methods without task decomposition. One possible explanation for this result is that novices possess relatively lower levels of knowledge and cognitive abilities than professional designers [52]. Therefore, the conceptual knowledge of professional designers may overlap with some of the ideas generated by the FBS-based method which provides unsatisfactory assistance in the professional's conceptual ideation. In contrast, for novice designers, the FBS-based method is more likely to generate unexpected outputs beyond their knowledge base, leading to more exploration of new ideas. Experts typically solve problems in their domain more effectively than novices [64]. However, it is noteworthy that the final scores of novices in the G2 group were 0.08 points higher than those of professionals in our experiment. One possible explanation is that a transition of the standard FBS task decomposition and representation means resulted in novices outperforming experts. The results are consistent with Wiley's research [65], which emphasizes that experts' knowledge may limit themselves to a certain region of the search space when a shift from the standard means of representation is required.

6.3 Function–Behavior–Structure-Guided Prompting Design. In this study, the FBS ontology-based prompt templates and specification signifiers are used in each mapping stage in experiments. The prompt templates are handcrafted in a semi-constrained form, allowing the user to enter only the corresponding FBS ontological words [52]. It is unclear how the degree of constraint in the template or the type of prompt pattern affects the quality of the generated content. Future research can analyze the performance of the FBS decomposition-based framework by exploring different forms and strategies of prompt templates. Specification signifiers constrained the scope of generated content through three keywords of FBS mapping. "Function," "afford," and "structure" were identified subjectively when organizing experiments. Although the keywords provide relatively accurate information from a semantic perspective, it remains unclear whether other words or phrases used as specification signifiers impact the generated results [66]. Furthermore, the selection of keywords that produce coherent and effective outputs is usually empirical, with potential keywords identified through continuous practice [67]. A future research direction could explore the list of valid specification signifiers for the FBS-based AI-aided approach, assessing the strengths and weaknesses of these signifiers.

In this study, the "function–structure" stimuli were derived from natural language-based textual results obtained after reasoning with ChatGPT and selected by human designer evaluation. However, the evaluation between Be and Bs may lead to human bias and knowledge limitations. Although good design may depend on human preferences, avoiding overly radical choices should be considered. A future study can be considered on how to introduce certain metrics to increase the rationality of "function–structure" selection. In addition, considering that designers in real design tasks often receive multi-modal stimuli in natural language and visual images [68], this type of stimuli with single textual representation generated by LLMs may limit the designer's ideation ability. Previous research has shown that integrated stimuli of visuals and text can provide a more comprehensive understanding of concepts [69]. Future research will examine how text-image multi-modal interactions influence designers' decisions and the overall effectiveness of this approach.

6.4 Discussion of Experimental Settings. In this study, the sports rehabilitation device was chosen as the design problem for the comparison experiment. While this is a typical design case with relative complexity in the field of mechanical design, it may affect the reliability of the evaluation results and the adaptability of our approach. In addition, this design problem is limited to professional equipment in the medical and healthcare fields. Future work could investigate the feasibility of our approach in various engineering and design scenarios.

In our evaluation, the weights assigned to the five metrics of concept scoring were consistent. However, the importance of different indicators may vary across different conceptual design scenarios. For example, some studies may focus more on novelty in the conceptual design process [70]. Future research will consider the impact of varying these weights on our method. Additionally, there may be other metrics that better measure conceptual design performance, with concept scoring being just one of them. Therefore, more comprehensive research into conceptual design measurement is needed.

6.5 What Happens When Large Language Model Become More Powerful. The overall score for G1 was 1.37% higher than that for G3. One possible reason is that the model, fine-tuned with a specialized design database, is more likely to capture latent design-related knowledge. This also confirms that the knowledge-based approach using fixed prompts can perform conceptual design tasks at a reasonable level of competence [41]. However, using design knowledge to fine-tune the model seems to offer limited enhancement to the capabilities of AI-aided conceptual design. It is reasonable to believe that as LLMs become more powerful, conceptual design may no longer require design knowledge for fine-tuning. In addition, conceptual design is a task with creativity [71]. More powerful LLMs in the future may still require the design science method and human intervention to accomplish the task. With existing machine learning mechanisms, LLMs are still essentially probabilistic inference models that aim to return the most desirable outcomes [72]. However, good design often combines unusualness and rationality [73], a balance AI models find difficult to achieve.

7 Conclusion

In this study, we present a generative concept design approach based on the FBS task-decomposition model. The proposed approach uses function, behavior, and structure-related prompt templates to decompose design problems and guide LLM-aided ideation in three steps. The output of each step becomes the input for the next, thereby aggregating gains per step and embedding human designers' selection at each stage. Prompt templates and specification signifiers enhance the controllability of the model's

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domain-specific tasks, guiding the LLM to generate more reasonable design concepts. Comparative experiments demonstrate that the proposed task-decomposed AI-aided process for generative conceptual design effectively and comprehensively assists humans in complex tasks by leveraging the strengths of LLMs and the FBS model.

To the best of our knowledge, this study is the first attempt to integrate a task-decomposed design method to guide LLMs in conceptual design. Exploring the opportunities this approach offers for engineering design research is critical. By demonstrating the approach's significant potential for human-computer generative conceptual design, we hope to inspire further research in the DTM field and AI-aided innovation.

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Conflict of Interest

There are no conflicts of interest.

Data Availability Statement

The datasets generated and supporting the findings of this article are obtainable from the corresponding author upon reasonable request.

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